

CO2 -ASSESSMENT TOOL 3

COURSE NAME: **CSA16 COURSE CODE: Data Warehousing and Data Mining**

SCENARIO BASED QUESTIONS

Q1. Data Preprocessing in Retail Analytics (*L3, L4*)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age**, **Monthly_Spend**, and **Purchase_Frequency**
 - Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
 - Compute the imputed values and justify the selected method
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B. Outlier Detection and Treatment

- Detect outliers in **Age** and **Monthly_Spend** using the **Z-score method**
 - Identify records containing extreme values (e.g., Age = 105, Spend = 500)
 - Apply suitable treatment methods (**capping** / **transformation** / **removal**) and justify
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C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male** / **Female**)

A. Handling Missing Values

i. Missing Values

Attribute	Missing Record
Age	C2
Monthly_Spend	C3
Purchase_Frequency	C5

ii. Two Imputation Techniques

Method 1: Median Imputation

- Age values = 23, 31, 105, 27, 40, 36, 42
 - Median = 36
- Monthly Spend = 20, 35, 500, 25, 45, 38, 300
 - Median = 38
- Purchase Frequency = 5, 8, 10, 50, 12, 9, 45
 - Median = 10

Imputed values:

- C2 Age = 36
- C3 Monthly Spend = 38
- C5 Purchase Frequency = 10

iii. Justification

Median imputation is preferred because it is simple and resistant to outliers like Age = 105 and Spend = 500.

B. Outlier Detection

i. Z-score Method

Outliers detected:

- Age = 105
- Monthly Spend = 500

ii. Extreme Records

Customer Outlier

C4 Age =105, Spend =500

C8 Spend =300 (possible high value)

iii. Treatment

- Cap Age to maximum acceptable value (e.g., 60).
- Cap Monthly Spend to upper percentile.
- Alternatively remove extreme records if caused by entry errors.

C. Categorical Standardization

Original values

- M
- Male
- male
- F
- Female
- FEMALE

Standardized
Original Standard
M Male
Male Male
male Male
F Female
Female Female
FEMALE Female

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
 - ii. Interpret whether the relationship is **positive or negative**
 - iii. Explain how covariance helps in identifying feature relationships
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B. Feature Selection

- i. Identify redundant attributes using **correlation or domain knowledge**
 - ii. Select an optimal subset of features for modeling
 - iii. Justify your selection
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C. Data Transformation

- i. Standardize the dataset (excluding **Cust_ID** and **Customer_Class**)
- ii. Explain why normalization/standardization is required

A. Covariance

Income and Purchase Value both increase together.

Hence,

Covariance > 0 (Positive).

Interpretation:

Higher income customers generally make higher purchases.

Covariance helps identify relationships between variables before model building.

B. Feature Selection

Possible redundant features:

- Income
- Average Balance
- EMI

Selected features:

- Age
- Income
- Purchase Value
- Credit Score
- Orders

Justification:

These features provide sufficient information with less redundancy.

C. Data Transformation

Standardize all numerical attributes except:

- Cust_ID
- Customer_Class

Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Why?

- Removes scale differences
 - Improves ML accuracy
 - Faster convergence
 - Prevents one feature dominating others
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Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student ID	Study Hours	Test Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
- Calculate bin width and define intervals

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
- Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

A. Equal Width Binning

Study Hours

Range

$$11 - 2 = 9$$

Bin Width

$$9/3 = 3$$

Bins

- Bin1: 2–5
- Bin2: 5–8
- Bin3: 8–11

Test Score

Range

$$100 - 40 = 60$$

Bin Width

$$60/3 = 20$$

Bins

- 40–60
 - 60–80
 - 80–100
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B. Equal Frequency Binning

10 records $\div$ 3 bins $\approx$ 3–4 records

Study Hours

Bin 1

2,3,4

Bin 2

5,6,7

Bin 3

8,9,10,11

Test Score

Bin 1

40,50,55

Bin 2

65,75,85

Bin 3

90,95,98,100

C. Concept Hierarchy

Study Hours

- Low $\rightarrow$ 2–4
- Medium $\rightarrow$ 5–7
- High $\rightarrow$ 8–11

Test Score

- Poor $\rightarrow$ 40–59
- Average $\rightarrow$ 60–79
- Excellent $\rightarrow$ 80–100

Q 4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

Data

200, 300, 400, 600, 1000

(a) Min-Max Normalization

Formula

$$\frac{x-200}{1000-200}$$

Value Normalized

200 0.000

Value Normalized

300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-Score Normalization

Mean

$$\mu = 500$$

Standard Deviation

≈ 282.84

Value Z-score

200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Q 5. Data Transformation in Online Shopping Behavior Analysis (*L3, L4, L5*)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer ID	Visit Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
- Assign each value to its respective bin
- Analyse how effectively this method groups short and long visit durations

B. Z-Score Normalization

- Compute the **mean and standard deviation** of visit duration
- Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

- Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
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C. Min-Max Normalization

- Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

- Transform all values accordingly
- Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Visit Duration

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

A. Equal Frequency Binning

10 records ÷ 3 bins

Bin 1

15, 30, 45

Bin 2

60, 120, 300

Bin 3

600, 900, 1800, 3600

Analysis:

- Short visits grouped together.
 - Medium visits grouped together.
 - Long visits grouped together for customer segmentation.
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B. Z-score Normalization

Mean

$$\frac{15+30+45+60+120+300+600+900+1800+3600}{10} = 747$$

Standard deviation ≈ 1109.3

Formula

$$Z = \frac{x - 747}{1109.3}$$

Examples

- $15 \rightarrow -0.66$
- $300 \rightarrow -0.40$
- $900 \rightarrow 0.14$
- $1800 \rightarrow 0.95$
- $3600 \rightarrow 2.57$

Discussion:

Extreme values (1800 and 3600) receive high positive Z-scores, making outliers easy to identify.

C. Min-Max Normalization

Formula

$$\frac{x-15}{3600-15}$$

Duration Normalized

15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison:

- Min-Max Normalization: Scales all values to the range [0,1] but is sensitive to outliers.
- Z-Score Normalization: Centers data around mean (0) with unit variance and is better for identifying extreme values.